Project Title

College Picker

Project Summary

The College Picker is a web-based application designed to help students identify colleges that best match their personal, academic, and financial preferences. Many college search platforms exist, but they often provide limited filtering options or prioritize popularity and rankings over individual priorities. This application aims to solve that problem by giving users the ability to weigh factors such as tuition costs, location, program offerings, campus size, diversity, and scholarship availability. By leveraging data analysis, the College Picker offers personalized recommendations, helping students make more informed decisions and reducing the overwhelm of navigating hundreds of college options.

Creative Component

The creative component of this project will be a Pandas-based recommendation algorithm. It will be a unique way to transform the data, using such statistical concepts as normalization, to process multiple datasets, calculate weighted scores, and rank colleges accordingly. Additionally, depending on how much time we have, this algorithm could be augmented with simple machine learning techniques, such as linear regression, or an interactive dashboard built with Plotly or Altair that will allow users to explore college comparisons visually. In addition, we will also collect historical data from the past five years for dynamic metrics such as tuition, admission rate, and graduation rate. This data will be aggregated and aligned across years to enable trend analysis. For the top five colleges returned to the user, we will generate a dashboard that visualizes these multi-year trends through graphs and charts.

Usefulness

College Picker is useful because it transforms the overwhelming and confusing college search process into a structured, personalized decision making experience. Many students rely on generic rankings such as the U.S. News or marketing-driven platforms that do not account for individual priorities. Some students pay private educational consultants or agents to help them identify suitable colleges and seek for guides. We all have similar experiences and understand the difficulties students face when choosing a suitable college.

The functionalities of College Picker includes:

Filter colleges by multiple factors such as tuition, location, enrollment size, and available programs,

Assign importance weights to each factor to personalize results,

View a ranked list of best-fit colleges,

Compare colleges side by side in an interactive dashboard (extension functionality).

Some websites like this already exist (Niche, Collegeboard, Colleges of Distinction), but have limited flexibility in weighting criteria or combining multiple datasets. College Picker allows for greater customization using weighted factors and interactive visualizations to see the impact of their preferences. Unlike the existing platforms that require account registration or collect personal information, College Picker remains transparent and accessible, providing immediate results without unnecessary barriers.

Realness

To ensure realness and create an application that has practical use to students looking for colleges, we will use databases that allow the user to sort by important factors, such as price, location, religious affiliation, and average financial aid. The College ScoreCard API contains the aforementioned information about colleges, and the data can be saved into a CSV. It has several thousand rows with more than 60 columns. Another database, the IPEDS data center does overlap with the College ScoreCard, but it has more details for some features, especially student financial aid. We can get extra data as needed in CSV or xlsx format and join it with the College ScoreCard. It has nearly 6 thousand rows in the student financial aid dataset, with over 600 columns.

We also have some other potential sources: Peterson's Data, CollegeAI API.

As this will be a real application, user data (login information and survey responses) will also be saved securely in the database if the user chooses to sign up. We can generate mock data for it using AI, however we will still make it secure as in theory it would store actual user information. Storing user information adds to the risk of an attack, so we must create a database with a reduced attack surface, and defend against common security concerns, such as SQL Injection. There are other security concerns such as Distributed Denial of Service (DDoS) attacks, but as this is a project that heavily focuses on the database aspect, it is best to spend time and effort on the ones related to the database and SQL, and only think about the others given extra time.

Functionality

This website provides an interactive college recommendation system that helps students efficiently identify the best college matches based on their personal preferences and academic profile. Users can either use the system immediately without creating an account, or create an account to save their data for future access.

1. Guest Mode (No Login Required)
   Users may access the recommendation tool without creating an account
   a. What a guest user can do:
      i. Fill out a preference form including:
         1. Preferred state or region
         2. School size (small, medium, large)
         3. Religious affiliation preference
         4. Tuition affordability range
         5. GPA and/or test scores
         6. Other criteria (public/private, urban/rural, etc.)
         7. Relative importance (weights) of each factor
      ii. Submit the form to generate recommendations
      iii. Receive:
         1. Top 5 college matches
         2. Key statistics for each college
         3. A compatibility score
   b. Data Behavior:
      i. Preference data is processed temporarily
      ii. Data is **not stored in the database**
      iii. On page refresh or session end, all user input is lost

This mode allows users to quickly experiment with the tool without commitment. This will also set us apart from other college picking websites.

2. Registered User Mode (Login + Data Persistence)
   Users may optionally create an account to save their preferences and results.

a. Account Features:
    i. Create account (insert user credentials into database)
    ii. Log in / log out
    iii. Update account information
    iv. Delete account
b. Preference Management
    i. Users submit the same preference form as guest users
    ii. On first submission:
        1. Preferences are **inserted into the database**
    iii. On subsequent submissions
        1. Preferences are **updated**
    iv. Users can revisit their saved preferences at any time
c. Saved Data Includes:
    i. User profile information
    ii. Saved preference weights
    iii. Most recent recommendation results (optional enhancement)

This allows users to track changes, refine preferences, and return later without re-entering information.

3. Recommendation Engine

The backend stores structured college data (location, tuition, size, acceptance, etc.)

When a user submits preferences:

a. The system retrieves relevant college data
b. A Pandas-based scoring algorithm:
    i. Applies user defined weights
    ii. Calculates compatibility score
    iii. Ranks colleges accordingly
c. System returns the top 5 or top X best matches

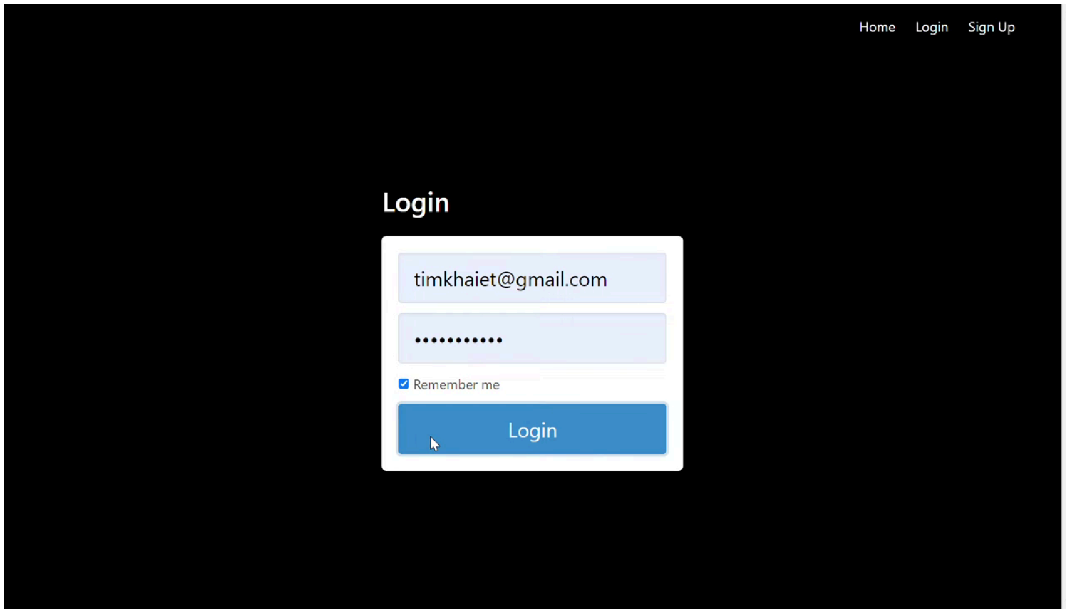
4. College Data Management
    a. College data is stored in the database, with 1 table for each data source
    b. Data can be periodically updated via API calls
5. Optional Interactive Dashboard (If Implemented)
    a. Compare selected colleges visually
    b. Filter by cost, size, acceptance rate, or other metrics
    c. View dynamic charts (scatterplots, bar charts, radar charts)
    d. Adjust weights and see real-time ranking updates
6. Design Rationale

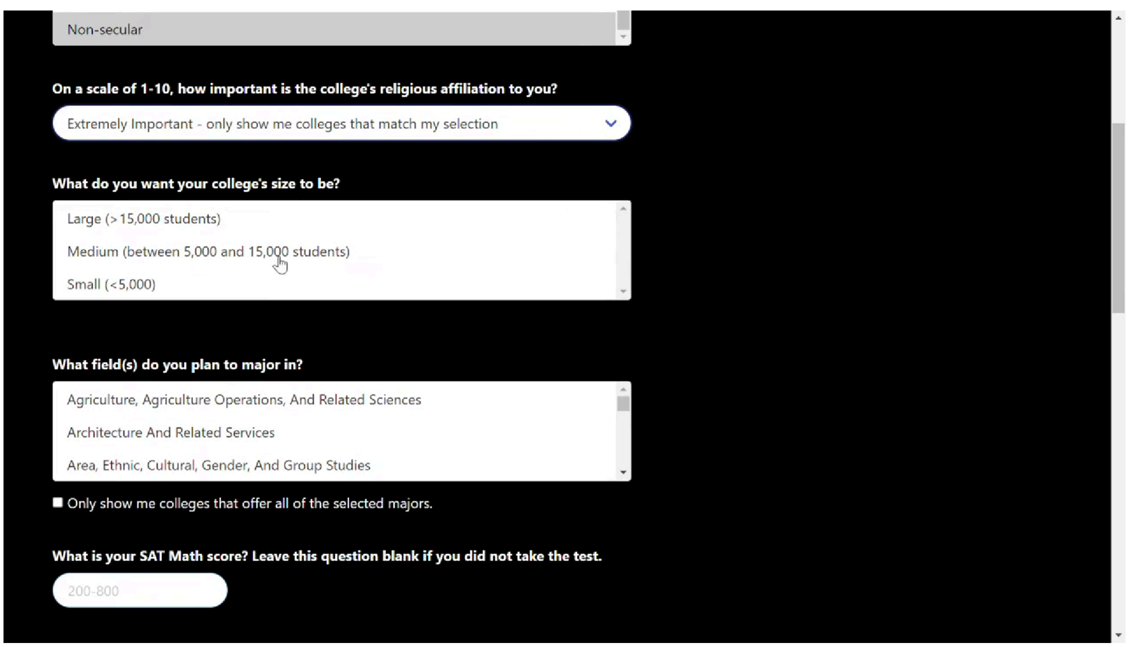
    By making login optional, the website balances accessibility and functionality

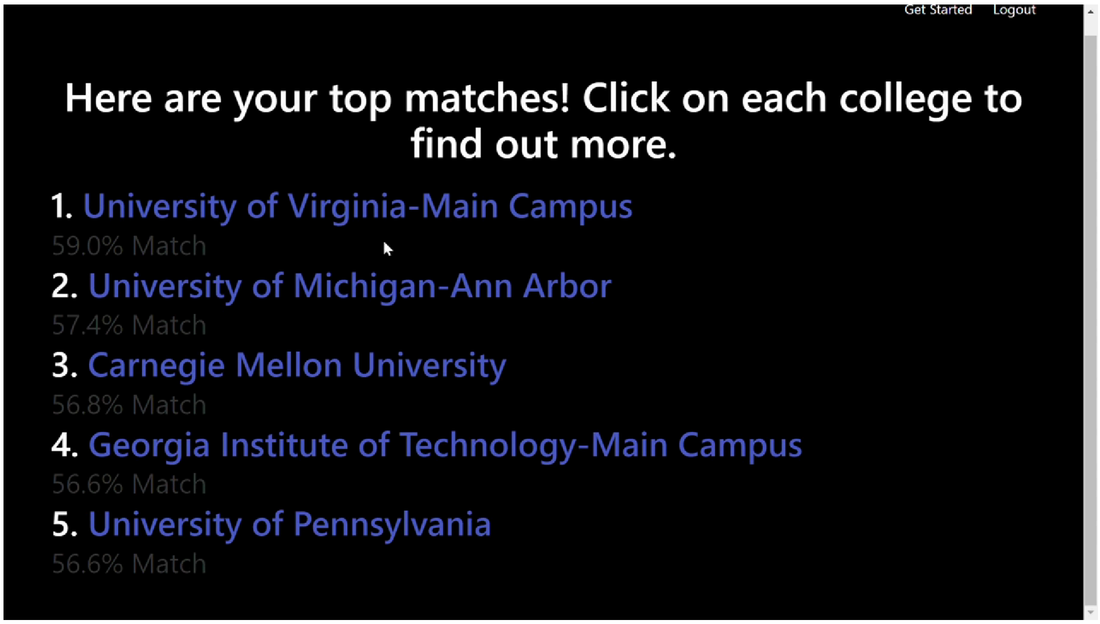
    a. Guest mode lowers the barrier to entry and encourages experimentation
    b. Account mode supports long-term planning and data persistence

This design makes the system both beginner-friendly and powerful for serious users refining their college search over time

UI Mock-Up


Home
Login
Sign Up
Login
timkhaiet@gmail.com
Remember me
Login


Non-secular
On a scale of 1-10, how important is the college's religious affiliation to you?
Extremely Important - only show me colleges that match my selection
What do you want your college's size to be?
Large (>15,000 students)
Medium (between 5,000 and 15,000 students)
Small (<5,000)
What field(s) do you plan to major in?
Agriculture, Agriculture Operations, And Related Sciences
Architecture And Related Services
Area, Ethnic, Cultural, Gender, And Group Studies
Only show me colleges that offer all of the selected majors.
What is your SAT Math score? Leave this question blank if you did not take the test.
200-800



Project Work Distribution

Task	Team Member Responsible
Data Collection & Cleaning (College Scorecard, IPEDS)	Artem
Backend: Pandas Recommendation Algorithm	Artem
Frontend: UI/UX, Web Interface & Forms	Zhenzhen
Interactive Dashboard & Visualizations	Kyle
API Integration & Data Updates, Database Management	Kyle
Sign In/User Authorization & Security Handling	Jimmy
Testing & Debugging	All members
Documentation & Presentation	All members